COMPREHENSIVE MEP DESIGN BASIS REPORT (DBR) ? SUPER BLOCK E:

1. METEOROLOGICAL DESIGN CONDITIONS (AMARAVATI CAPITAL REGION):
   ? Summer Design Ambient: 44.0°C Dry Bulb / 28.0°C Wet Bulb; Daily Range: 12.0°C.
   ? Monsoon Design Ambient: 34.0°C Dry Bulb / 29.0°C Wet Bulb.
   ? Indoor Design Conditions: 23.5°C +/- 1.0°C Dry Bulb; 55% +/- 5% Relative Humidity.
2. COOLING LOAD ESTIMATION:
   ? Total conditioned gross floor area: 68,500 m2 across Assembly Hall, Council Hall, Secretariats, and Courtyards.
   ? Total estimated block cooling load: 3,200 TR (Including 15% safety factor and fresh air enthalpy recovery).
3. ELECTRICAL CONNECTED LOAD & DEMAND:
   ? Total connected electrical load: 5,500 kVA; Anticipated maximum demand: 4,100 kVA (Diversity factor = 0.75).
   ? Backup power requirement: 100% emergency standby back-up via 2x 1500 kVA + 1x 1000 kVA DG sets.
4. WATER SUPPLY & SUSTAINABILITY (GRIHA 5-STAR TARGET):
   ? Total water requirement: 220 KLD; Fresh potable water: 95 KLD; Recycled treated sewage water for flushing/HVAC: 125 KLD.

Building Sector	Conditioned Area m2	Cooling Load TR	Power kVA	Water Demand KLD
Sector A: Legislative Chamber	18,500 m2	980 TR	1,650 kVA	65 KLD
Sector B: Council Chamber	14,200 m2	720 TR	1,200 kVA	45 KLD
Sector C: Speaker & Cabinet	16,800 m2	780 TR	1,350 kVA	55 KLD
Sector D: Media, Press & Visitors	19,000 m2	720 TR	1,300 kVA	55 KLD
Total Complex Gross	68,500 m2	3,200 TR	5,500 kVA	220 KLD